Tribhuvan University

Faculty of Humanities and Social Science

Semester: IV

Subject: Numerical Methods

2021 Batch

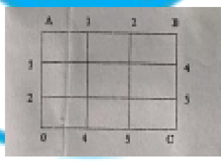
Group B

Attempt any SIX questions

2. Explain absolute and relative error. Find the relative error of number 5.6 if both of its digits are correct.
3. On what type of equations Newton's method can be applicable-Justify.
4. Solve the following equations by using Gauss-Jordan method.
   2x+3y+4z=5   3x+4y+5z=6   4x+5y+6z=7
5. Use the Romberg method to get an improved estimate of the integral from x = 1.8 to x = 3.4 from the data in the table with h =0.4.

X:	1.6	1.8	2.0	2.2	2.4	2.6	2.8	3.0	3.2	3.4	3.6	3.8
Y:	4.953	6.050	7.389	9.025	11.023	13.464	16.445	20.056	24.533	29.964	36.598	44.701

6. Write a program to compute integral $\int_0^{\pi/2} \sqrt{sinx}\, dx$ Simpson's 1/3 rule.
7. Using Runge-Kutta method of 4th order solve the following equation taking each step h = 0.1 $\frac{dy}{dx} = \frac{4x}{y} - xy$ given y(0) = 3 3.calculate y at x = 0.1 and 0.2.
8. Solve the lapiace equation Uxx + Uyy = 0 for the following square mesh with the boundary values as shown in the figure below:



Group C

Attempt any TWO questions.

9. Solve the given set of linear equations using Dolittle LU decomposition method:
$$3x_1 + 2x_2 + x_3 = 10$$
$$2x_1 + 3x_2 + 2x_3 = 14$$
$$3x_1 + 2x_2 + 3x_3 = 14$$
10. Define initial value problems and final value problems. Using heun's method, find value of y when x=0.3 given that $\frac{dy}{dx} = x + y$ and y=1 when x=0.
11. How can x we use Laterpolation techniques (methods) to approximate the value of the integral for the functions whose antiderivative can't be found? Explain. Write a program to solve sin x - 2x + 1 = 0 using Bisection method.

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2020 Batch

Group B

Attempt any SIX questions

2. Define error. Explain the Taxonomy Errors.
3. Estimate the value of sin θ at θ = 45° using Newton's backward difference formula from the following set of data.

θ	10	20	30	40	50	60
Sin θ	0.1736	0.3420	0.5000	0.6428	0.7660	0.8660

4. Write an algorithm and program to calculate integration using Trapezoidal rule.
5. What is the form of resultant matrix using Gauss-Jordan method? Solve the following system of equations using Gauss-Jordan Method.]

   x + 2y-3z=4

   2x+4y - 6z = 8

   x-2y+5z=4
6. Define ordinary differential equation. Use the fourth order Runge-Kutta method to estimate y(0.4) of the equation $\frac{dy}{dx} = x^2 + y^2$ with y(0) = 0 assuming that h = 0.2.
7. 7. Solve for the steady-state temperatures in a rectangular plate of 8cm x 10cm, if one 10cm side is held at 50°C, and the other 10cm side is held at 30°C and other two sides are held at 10°C.Assume grids of size 2cm x 2cm.
8. Write a short note on (Any Two):

   a) Partial Differential Equations a) Linear Interpretation b) Boundary value problems

Group C

Attempt any TWO questions.

9. Write an algorithm and program to compute the root of nonlinear equation using Newton - Raphson method.
10. a) Fit a straight line to the following set of data using Least Square Regression.

    b) Apply the factorization method (any) to solve the equations:

    3x+2y+7z=4

    2x +3y +z=5

    3x +4y +z=7
11. a) Write and implement an algorithm to solve the system of linear equations using Gauss-Seidel method with suitable numerical example.

    b) Write a program to solve the ordinary differential equations using Heun's method.

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2019 Batch

Group B

Attempt any SIX questions

2. If the true value of π is 3.1415926 and its approximate value is given by 3.1428571. Find the absolute and relative errors.
3. What are the advantages of Lagrange's formula over Newton's formula? When Newton's forward interpolation formula is used?
4. Find the polynomial that interpolates f(0) = 0, f(1) =1, f(2) = 0, f(3) = 1, f(4) = 0 using Newton divided differences. Use the Newton table to generate the necessary divided differences
5. Using the trapezium rule, evaluate the integral $I = \int_0^1 \frac{dx}{x^2+6x+10}$ with 2 and 4 sub intervals. Compare with the exact solution. Comment on the magnitudes of the errors obtained.
6. For solving a linear system, compare Gauss elimination method and Gauss Jordan method. Why Guass siedel method is better than Jacobi's iterative method?
7. Using Euler's method find y (0.2) from $\frac{dy}{dx} = x + y$, y (0) =1 with h -0.2.
8. Solve the Poisson equation $\Delta^2 f = 2x^2y^2$ over the square domain $0 \leq x \leq 3$ and $0 \leq y \leq 3$ with f = 0 on the boundary and h= 1.

Group C

Attempt any TWO questions.

9. When would we not use N-R method? Find the root of the equation $x^3 - 4x + 1 = 0$, lying in (0, 1) Using Bisection method performing 10 iterations.
10. Find the dominant eigen value of A=$\begin{pmatrix} 2 & 3 \\ 5 & 4 \end{pmatrix}$ by power method. Apply Gauss seidel method to solve system of equations.
    6x1 – 2x2+x3 = 11; - 2x1 +7x2+2x3=5; x1 +2x2 - 5x3 =- 1, with the initial vector of (0, 0, 0).
11. Prepare the multi-step methods available for solving ordinary differential equation, Evaluate the value of y at x=0.1 and 0.2to 4 decimal places given $\frac{dy}{dx} = x^2y - 1$ y(0)=1, using Taylor series method.

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2018 Batch

Group B

Attempt any SIX questions

2. Compute the root of equation $x^2 - 5x + 6 = 0$ in the vicinity of x = 5 using Newton-Raphson method.
3. Estimate the value of ln(2.5) using Newton-Gregory forward difference formula given the following data.

x	1.0	2.0	3.0	4.0
Ln(x)	0.0	0.6931	1.0986	1.3863

4. Write an algorithm to compute integration using Simpson's 1/3 rule. Evaluate integration of $\int_0^2 (3x^2 + 2x - 5)dx$ using Simpson's 1/3 rule. [2.5 + 2.5]

   What is meant by ill-conditioned system? Find the Cholesky decomposition of the following matrix. [5]

$$\begin{bmatrix} 4 & 1 & 1 \\ 1 & 5 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

6. Use the classical RK method to estimate y(0.4) of the equation $\frac{dy}{dx} = x^2 + y^2$ *with* $y(0) = 0$ assume h = 0.2. [5]
7. Solve the Poisson equation $\nabla^2 f = 2x^2y^2$ over the square domain $0 \le x \le 3$ and $0 \le y \le 3$ with f = 0 on the boundary and h = 1. [5]
8. Write a short note on (Any Two): [2.5 + 2.5]
   a) Types of Errors
   b) Convergent Methods
   c) PDE

Group C

Attempt any TWO questions. [2×10 = 20]

9. Write an algorithm and program to compute root of nonlinear equation using bisection method. [4 + 6]
10. Given then data points:

i	0	1	2
x_i	9	16	25
f_i	3	4	5

Estimate the function value of f at x = 14 using cubic splines. [10]

11. a) Solve the following system of equation by Gauss-Seidel method. [5]

$$2x - 7y - 10z = -17$$
$$5x + y + 3z = 14$$
$$X + 10y + 9z = 7$$

b) Write an expansion of Taylor's Theorem. Solve the equation $y'(x) = x^2 + y^2$, y(0) = 1 for x = 0.5 using Taylor method. [2.5 + 2.5]

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2017 Batch

Group B

Attempt any SIX questions

2. Why is the study of errors important to a computational scientist? Differentiate between inherent and numerical errors.
3. Find the root of equation $x^2 - 4x - 10 = 0$ using bisection method where root lies between 5 and 6.
4. Find the square root of 3.5 using second order Lagrange interpolation polynomial using the following data table.

x	1	2	3	4	5
f(x)	1	1.4142	1.7321	2	2.2361

5. Write a program to calculate the integral using Trapezoidal Rule.
6. Solve the following set of equations using Gauss-Jordan Method.
   3x-5y+2z=15
   4x-y +z=2
   x-3y+7z=22
7. Use classical Runge-Kutta method to estimate y(0.2) when y'(x) = $x^2 + y^2$ with y(0) = 0 and h =0.2.
8. Solve the Poisson equation $\nabla^2 f = 2x^2y^2$ over the square domain $0 \le x \le 3$ and $0 \le y \le 3$ with f=0 on the boundary and h = 1.

Group C

Attempt any TWO questions.

9. a) Factorize the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 22 \\ 3 & 22 & 82 \end{bmatrix}$ Using Cholesky's method.

   b) Locate the root of equation $x^2 + x - 2$ using the fixed point method.
10. a) Fit a straight line to the following set of data points.

x	1	3	4	6	8	9
y	1	3	4	4	5	7

   b) Differentiate between ordinary and partial differential equations with their applications.
11. Given the data points:

i	0	1	2
xi	2	3	4
fi	4	9	16

Estimate the function value f at x = 2.5 using cubic splines.